

Drithya: Automated Compliance System

Smart India Hackathon 2026

Project Presentation

High-level overview of the compliance software, its benefits, and the impact of automation on supply chain efficiency.

1. The Problem: Manual Compliance Checks

The Legal Metrology (Packaged Commodities) Rules, 2011, mandate strict labeling requirements. Currently:

- **Manual Audits:** Enforcement officials and manufacturers rely on manual verification.
- **High Error Rates:** Human error leads to non-compliance, heavy fines, and product recalls.
- **Supply Chain Bottlenecks:** Manual checks slow down product distribution and inventory cycles.

2. Our Solution: Drithya

An AI-powered, deterministic compliance engine that automates label verification.

- **Instant Validation:** Replaces hours of manual checks with seconds of automated verification.
- **Deterministic Logic:** Translates LMPC rules into hard-coded, rule-based algorithms.
- **End-to-End Workflow:** Supports live scanning, image uploads, and e-commerce listing verification.

3. Key Features & Architecture

- **Intelligent OCR Pipeline:** Extracts text, dates, weights, and symbols directly from product packaging.
- **Rule Engine:** Verifies mandatory declarations (MRP, Expiry, Net Quantity, Manufacturer details).
- **Automated Reporting:** Generates detailed, deterministic audit trails for every scan.
- **Evidence Management:** Stores compliance logs securely for legal and regulatory audits.

4. Impact on Supply Chain Efficiency

- **Reduced Bottlenecks:** Faster compliance clearance accelerates time-to-market.
- **Cost Savings:** Prevents expensive recalls and fines due to mislabeling.
- **Proactive Compliance:** Empowers manufacturers to catch errors *before* products hit the shelves.
- **Transparency:** Provides a single source of truth for both manufacturers and enforcement officials.

5. Conclusion & Future Scope

Drithya bridges the gap between regulatory requirements and modern supply chain speeds.

By translating complex legal rules into programmable logic, we ensure accurate, rule-based identification of product label violations.

Next Steps:

- Integration with Government APIs
- Expansion to regional languages
- Real-time factory-line deployment

Thank You!